

A Framework for Spacetime from Weighted Histories

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Abstract

We formulate a research framework for investigating whether spacetime and quantum fields can arise from a common underlying structure. Candidate histories are finite relational structures considered up to isomorphism. Admissibility conditions specify which histories are possible, while a separate weighting specifies how alternatives contribute to predictions. To allow interference, the weighting is a strongly positive decoherence functional rather than an ordinary probability distribution over individual histories. Probabilities are assigned to decoherent families of alternatives. This quantum structure is an assumption, not a derivation from combinatorics. No external growth time is introduced. Physical time is instead a proposed relation between internal clock readings and other records. The central question is whether suitable admissibility conditions and weights support effective geometry and matter dynamics that remain stable under changes in microscopic details. We specify the finite framework and the reconstruction tasks, but do not supply a physical model or establish such a regime. The selection of the weights, recovery of general relativity and quantum field theory, and definition of a horizon entropy remain open.

1 Introduction

General relativity describes a dynamical spacetime geometry, while quantum field theory describes quantum matter and its interactions on a specified spacetime background. The aim of this work is to investigate a structure from which both descriptions could be recovered in appropriate regimes. The underlying objects need not themselves be points of spacetime, and their mathematical description need not begin with a metric or an external time parameter.

The proposal concerns a space of admissible histories together with weights that determine how alternatives contribute to observations. A history is a relational structure, not initially a trajectory through an already defined space. Admissibility specifies which relations can coexist. Weighting is a separate part of the model, since knowing which structures are possible does not by itself determine their relative contributions. Quantum interference requires more than an ordinary probability distribution over alternatives. We therefore use a finite decoherence-functional description, drawing on the histories-based treatment of quantum theory [7]. This imports a quantum probability framework. It does not explain its origin.

The physical question is whether such a model can support persistent organisation with a useful spacetime description. A successful account would explain why the same effective laws survive a range of changes in microscopic details. It would also have to distinguish structures that are merely admissible from regimes that have appreciable probability within an appropriate decoherent family. Neither persistence nor probability is defined by resemblance to the physics one hopes to recover.

In this formulation there is no external process that traverses the space of histories. A calculation that enumerates a history is not itself a physical clock. Time would have to be recovered from correlations between internal records, in the spirit of relational approaches [5]. This leaves the construction of clocks and the origin of an arrow of time as separate problems. Likewise, cosmological expansion would concern an effective metric relative to such clocks, not the number of objects introduced by an enumeration algorithm.

Relational structures, histories, and inference principles already occur in causal set theory [1, 6, 8], rewriting approaches [3], and entropic dynamics [2]. These precedents do not establish the present proposal, and no claim of novelty for these ingredients is made here. The required comparison with those programmes remains part of the work.

This is a framework paper, not a derivation of a physical theory. The methodology separates admissibility, quantum weighting, records, and effective geometry so that assumptions in one part cannot be presented as results of another. It supplies definitions and identifies the missing constructions. No particular relational signature, weighting law, or continuum limit is shown to reproduce general relativity or quantum field theory.

2 The framework

2.1 Scope and assumptions

The starting point is a space of admissible histories, not a growing spacetime. A history specifies relations between elementary constituents. It need not come with a preferred sequence in which those constituents were created. The description is pre-geometric in the limited sense that no metric, physical dimension, or manifold structure is assigned to it. Whether it admits a spacetime interpretation is a question to be answered, not a property conferred by calling it a history.

Three choices have to be kept distinct. Admissibility determines which histories are allowed. Weighting determines how alternatives contribute to predictions. A description of records determines which questions those predictions concern. The framework does not yet derive any of these choices from the others.

We work at finite size so that the definitions do not presuppose a measure on an unspecified collection of infinite objects. Removing that restriction will require a separate limiting construction. No conservation of homotopy type, growth rule, or global fitness function is imposed. Such conditions could be added to a particular model, but each would require its own justification.

2.2 Admissible histories

Definition 2.1 (Candidate history). Fix a finite set of labels L and a finite relational signature Σ , whose symbols have specified finite arities. A candidate history H consists of a finite nonempty set E_H , a label map $\ell_H : E_H \rightarrow L$, and a relation $R_H \subseteq E_H^k$ for every k -ary symbol $R \in \Sigma$. An isomorphism of histories is a bijection of their underlying sets that preserves labels and all relations.

The elements of E_H are not assumed to be spacetime points or measurement events. A directed relation can be included, but its interpretation as physical causality would require further work. If its transitive closure is required to be a partial order, acyclicity is an additional constraint. It does not follow from the definition of a relational structure.

Definition 2.2 (Admissibility data). Admissibility data $\mathcal{A}$ consist of L , Σ , and a specified collection of isomorphism-invariant constraints on candidate histories. For a positive integer N , let

$$\Omega_N(\mathcal{A}) = \{H : H \text{ satisfies } \mathcal{A}, 1 \leq |E_H| \leq N\} / \cong.$$

We use only cutoffs for which this finite set is nonempty.

The cutoff N limits the size of a description. Increasing it is not identified with the passage of physical time. The phrase “all histories” will always mean all histories admitted by specified data $\mathcal{A}$, not all mathematical structures without restriction.

Locality also needs a definition. One available combinatorial notion uses the incidence graph whose vertices are elements of E_H , with two distinct elements adjacent when they occur together

in a relation tuple. A model can impose conditions on neighbourhoods of a specified finite radius in this graph. The graph, radius, and allowed patterns are then inputs. This does not establish locality with respect to a Lorentzian metric. Global conditions, including acyclicity, must be stated separately rather than described as consequences of local compatibility without an argument.

Passing to isomorphism classes removes arbitrary names for constituents. It does not identify distinct histories merely because they share a topology or some coarse description. Any further identification requires an explicit equivalence relation and an account of which observations it preserves.

[GAP: Choose a relational signature and admissibility constraints with an independent structural motivation. Specify the meaning of locality and any global restrictions. No particular choice is selected in this paper.]

2.3 Weights and interference

Admissibility alone does not supply probabilities. For fixed N and $\mathcal{A}$, write $\Omega_N = \Omega_N(\mathcal{A})$. A classical model could assign a probability $p_N(H)$ to each $H \in \Omega_N$. Such a model adds probabilities for disjoint alternatives and has no interference between them.

To leave room for interference, we instead introduce the following quantum weighting. This is an adopted mathematical structure, not a consequence of the preceding definitions [7].

Definition 2.3 (Finite quantum weighting). A weighting on Ω_N is a Hermitian positive semidefinite matrix K_N , indexed by pairs of histories, with normalization

$$\sum_{H, H' \in \Omega_N} K_N(H, H') = 1.$$

Its decoherence functional on pairs of subsets is

$$D_N(A, B) = \sum_{H \in A} \sum_{H' \in B} K_N(H, H').$$

The positive semidefinite condition on K_N imposes strong positivity.

The diagonal choice $K_N(H, H') = p_N(H)\delta_{H, H'}$ includes classical probability as a special case. Off-diagonal entries allow interference. For disjoint alternatives A and B ,

$$D_N(A \cup B, A \cup B) = D_N(A, A) + D_N(B, B) + 2 \operatorname{Re} D_N(A, B).$$

Consequently $D_N(A, A)$ is not, in general, an additive probability measure on all subsets of histories.

The pair $(\mathcal{A}, K_N)$ supplies finite kinematic and weighting data. A physical model would also have to distinguish its fixed dynamical law from preparation or boundary data on which K_N depends. Neither the admissibility constraints nor this matrix definition selects that law. Requiring weights to respect a specified local composition of structures could constrain it, but the composition and the constraint would need to be supplied explicitly.

[GAP: Construct a weighting prescription with a stated physical or structural justification. Establish its normalization, positivity, and compatibility with the proposed composition of histories. The complex weights and strong positivity are currently assumptions, not derived laws.]

2.4 Alternatives and records

Definition 2.4 (Decoherent family). A partition $\mathcal{P} = \{A_1, \dots, A_m\}$ of Ω_N is decoherent when $D_N(A_i, A_j) = 0$ for every $i \neq j$. Within that family, assign

$$p_N(A_i) = D_N(A_i, A_i).$$

These numbers are nonnegative and sum to one.

A partition specifies alternative answers to a question. It does not by itself describe a measuring device or a physical record. A record model would need to associate those alternatives with distinguishable structures inside the histories and explain how their correlations are retained. Identifying such structures is part of the reconstruction problem.

Different decoherent families need not admit a common decoherent refinement. The framework therefore does not assume one joint classical probability distribution for all possible measurement questions. The definition above uses exact decoherence. Any application using approximate decoherence must bound the interference terms and the resulting probability errors.

No operators, spectra, or measurement interactions have yet been recovered. In particular, the existence of this weighting formalism does not derive the eigenvalues of a quantum observable or the Born rule for a physical apparatus. Those claims would require a connection between the history description and a quantum theory of states, operations, and measurements.

[GAP: Define internal records and measurement alternatives. Show which families decohere and with what accuracy. Explain the relation to quantum observables without assigning independent classical values to all incompatible measurements.]

2.5 Time and invariance

The weighted space of histories is specified without a process that chooses one history after another. This removes the need for an external clock governing such a process. It does not establish that the physical universe is timeless, nor does it explain the occurrence of a definite measurement outcome.

A candidate internal clock would be a structure whose distinguishable readings correlate with other records. If the joint alternatives for a clock reading t and another record r form a decoherent partition, their conditional probabilities can be written

$$p_N(r \mid t) = \frac{D_N(A_r \cap A_t, A_r \cap A_t)}{D_N(A_t, A_t)}, \quad D_N(A_t, A_t) > 0.$$

This is a way to express relational predictions, not a derivation of dynamics [5]. A useful clock also requires ordered readings, appropriate correlations, and a regime in which it reliably parametrizes other changes. Conditional probabilities alone provide none of these.

An arrow of time would require an asymmetry in the relevant record correlations or boundary conditions. It is not supplied by the order in which a computer enumerates a history. Distinct microscopic histories can agree on clock readings, but that fact does not establish a second hidden time or a physical process outside the clock description.

Likewise, a fixed space of possibilities is not a conservation law. Relabelling invariance is built into Ω_N . Independence of predictions from a cutoff, a presentation, or an added equivalence relation is a different requirement. Each must specify the transformation and the predictions claimed to remain unchanged. Homotopy equivalence alone does not imply equality of geometry or physical observations.

[GAP: Construct an internal clock and identify its domain of validity. Determine which record correlations distinguish a temporal direction. No physical time observable or arrow of time has been derived.]

2.6 Effective geometry and matter

A spacetime reconstruction would assign geometric descriptions to suitable coarse alternatives, not identify every microscopic constituent with a spacetime point. It must specify what is being coarse-grained, which histories contribute to an alternative, and when a metric description is accurate. If a causal order is used, its origin and its relation to operational causality must be stated. A volume measure and physical scale cannot be inferred from the word “order.”

The target is a regime in which effective geometry and matter predictions come from the same admissibility and weighting data. A Lorentzian metric without gravitational dynamics would not recover general relativity. Random outcomes without interference, the appropriate observable algebra, and quantum correlations would not recover quantum field theory. Fitting these two descriptions independently would not establish their common origin.

Cosmological expansion would have to be expressed through an effective metric relative to internal clock readings. Neither an increase in the cutoff N nor a greater number of admissible histories is, by definition, an increase in spatial volume. No general principle requiring a closed system to settle into its lowest-energy configuration is assumed.

Entropy also depends on the question being asked. For a decoherent family, $-\sum_i p_N(A_i) \log p_N(A_i)$ is a Shannon entropy of those alternatives. It is not automatically thermodynamic entropy, entanglement entropy, or horizon entropy. Counting microscopic descriptions or their enumerations does not establish an area law.

[GAP: O1. Recover an effective Lorentzian geometry and its local Lorentz symmetry. Identify a regime in which gravitational and matter predictions recover general relativity and quantum field theory from the same model.]

[GAP: O2. Define a region, a horizon, its area, and the entropy associated with inaccessible records or degrees of freedom. Only then can an area-law comparison be posed.]

2.7 Robustness of effective laws

The proposed explanation of effective laws is that they might persist across a range of microscopic choices. This is a question about a family of models, not a property established by giving one model a complicated state space. It requires specified changes to admissibility or weighting and a way to compare the same coarse questions across those changes.

For example, suppose two models give probability distributions p_i and $\tilde{p}_i$ on matched coarse alternatives, each in a decoherent family. An operational tolerance can be expressed as

$$\frac{1}{2} \sum_i |p_i - \tilde{p}_i| \leq \epsilon.$$

A robustness claim must name the alternatives, the permitted microscopic changes, the tolerance, and the regime over which the bound is meant to hold. This comparison does not require an external physical time.

Persistence within a history is a separate property. It requires a way to recognize related records at different internal clock readings. Neither form of stability implies that a configuration is probable. In the interfering model, probability claims must refer to alternatives in a decoherent family rather than to an assumed classical distribution over all individual histories.

No global fitness function is needed for these questions. Conversely, neither local compatibility nor stability alone is claimed to force the observed laws. A model would have to exhibit the proposed regime and show which microscopic details its predictions do not depend on.

[GAP: Specify a family of microscopic models, matched coarse records, and allowed perturbations. Establish a nontrivial regime of robust predictions and distinguish this from persistence under an internal clock. No such regime has yet been demonstrated.]

2.8 Inference and limits

An inference principle could restrict the remaining choices, but its prior and constraints are part of the model. On the classical diagonal subclass, one possible finite problem is

$$\underset{p_N \in \mathcal{F}_N}{\text{maximize}} \left[- \sum_{H \in \Omega_N} p_N(H) \log \frac{p_N(H)}{\mu_N(H)} \right],$$

where μ_N is a specified strictly positive reference probability and $\mathcal{F}_N$ is a specified feasible set of probability distributions. Terms with $p_N(H) = 0$ are taken to be zero. This is a relative-entropy inference problem [4], not a derivation of the prior or the constraints. A uniform distribution over isomorphism classes is itself a choice and need not agree with counting labelled descriptions.

This classical problem does not determine off-diagonal quantum weights. Any inference principle for K_N must state its objective, admissible variations, and justification independently. No Gibbs form or unique quantum law is asserted here.

Finally, the family of finite models is not yet a theory of an infinite universe. A limiting proposal must specify how coarse records at different cutoffs correspond and whether their probabilities converge. It must also control dependence on the chosen regulator. Simply restricting a matrix from Ω_{N+1} to Ω_N need not preserve its normalization.

[GAP: Specify the inference data and a consistent limiting prescription. Complete the comparison with decoherent histories and quantum measure theory, causal set entropy and growth models, Wolfram model papers, entropic dynamics, network cosmology, and directed algebraic topology before making any claim of novelty.]

References

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